

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 05

Resource Loading & Leveling

Course	BECM 3101 - Construction and Project Management I
Primary lecture	5.0 - Resource Loading and Leveling (22 PDF slides)
Supporting source	5.1 - Resource Allocation and Resource Leveling (25-page reference chapter)
Question bank	Term-final examinations, 2016-2024 (25 pages)
Method	Every relevant question was mapped to the strongest primary slide, supporting slides and textbook-reference pages, regardless of the original Section A/B label.

RESULT

This lecture produced seven relevant question links across six examination years, worth 86 marks. The two dominant patterns are resource-loading problems and resource-leveling for a smoother resource profile. The 2024 paper adds the first full allocation-and-leveling numerical problem in this question bank.

Prepared as part of the cumulative lecture-by-lecture question-bank analysis.

1. Executive Summary

The primary deck covers resource definitions, resource loading, allocation constraints, usage calendars, AOA networks, load diagrams, resource leveling and the Minimum Moment Method. The second PDF is a reference-book chapter with fuller worked examples. Slide numbers in this report always use the PDF page count, including the title page as Slide 1.

22	25	7	5	2	86
Lecture slides	Reference pages	Question links	Exact matches	Direct matches	Mapped marks

Classification	Meaning
Exact	The lecture directly contains the requested definition, concept, diagram or answer structure.
Direct	The lecture contains the complete method or interpreting tool, but the exam uses new data or asks for an application not stated word-for-word.
Partial	Only part of the answer is available and another lecture or source is essential.

Numbering	Primary slides use PDF page numbers: for example, the resource-loading definition is PDF Slide 6 although the printed number on that slide is 5. Reference pages are also PDF page numbers, not the printed book-page numbers.
Section labels	All seven matches appeared under Section B in the original papers. They are included here because the user's current lecture grouping places this content in Section A and sections may be swapped.
No-hit years	No direct resource-loading or resource-leveling question was found in 2016, 2018 or 2020.

Highest-priority slide blocks

Priority	Slides/pages	Topic	Why it matters
1	6-8	Resource loading and allocation problems	Repeated in 2017 and 2022, forms the first part of 2021, and supports the 2024 chart question.
2	13-15	Resource leveling and smoother distribution	The same 12-mark theory question appeared in 2019 and 2023 and is also embedded in 2021.
3	10-12	AOA network and load diagrams	Essential for the simple-network explanation and for reading over-allocation or underutilization.
4	16-20	Minimum Moment Method	Central to the new 2024 allocation-and-leveling problem and a strong candidate for future numerical questions.
5	Ref. 8-19	Worked resource-cap examples	The reference chapter supplies complete before/after profiles and resource-limit cases that the lecture deck summarizes.

2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed meaning. 'Lecture slides' identifies the strongest answer source in the 5.0 deck. 'Ref. PDF' identifies supporting pages in the 5.1 reference file. 'QB p.' is the PDF page in the uploaded question bank.

Examinations 2017-2022

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Support	Ref. PDF	Coverage assessment
2017	Sec B 6(a)	15	5	Define resource loading and describe the problems associated with resource loading in the construction industry.	Exact	6-8	3-5	3	Slide 6 gives the definition and purpose of resource loading. Slides 7-8 explain why CPM/PERT alone is insufficient and distinguish time-limited and resource-limited allocation problems.
2019	Sec B 8(a)	12	11	What is resource leveling? How does it create a smoother distribution of resource usage? Explain with a simple network.	Exact	13-15	10-12, 16	3, 8-10	Slide 13 states the definition and smoothing purpose; Slides 14-15 show benefits and a before/after network. Slides 10-12 supply the network and load-profile basis.
2021	Sec B 8(a)	12	17	What is resource loading? How does resource leveling create a smoother distribution of resource usage? Explain with a simple network.	Exact	6, 13-15	10-12	3, 8-10	This mixed question is covered across the deck: resource loading on Slide 6, leveling on Slide 13, and the simple network/load-profile explanation on Slides 10-15.
2022	Sec B 8(a)	12	19	Define resource loading and describe the problems associated with resource loading in the construction industry.	Exact	6-8	3-5	3	This repeats the 2017 wording. The best answer path is Slide 6 followed by the allocation-problem discussion on Slides 7-8.

Examinations 2023-2024

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Support	Ref. PDF	Coverage assessment
2023	Sec B 6(a)	12	21	What is resource leveling? How does it create a smoother distribution of resource usage? Explain with a simple network.	Exact	13-15	10-12, 16	3, 8-10	This is essentially the 2019 question repeated. The definition, purpose, load profile and before/after network are all present.
2024	Sec B 5(a)	5	24	Describe how a resource loading chart helps project managers identify potential overallocation or underutilization of resources.	Direct	12	6, 9-11, 13	8-10	Slide 12 is the central load diagram. Slides 9-11 show the resource calendar and network. The reference example explicitly demonstrates a load spike above the available limit and the corrected leveled profile.
2024	Sec B 5(c)	18	24	Allocate and level resources for a nine-activity project with a maximum of 10 labour resources, without increasing the critical path (histogram excluded).	Direct	10-20	7-8, 21	7-22	The activity data are new, but the complete procedure is supported: network and load diagram (Slides 10-12), leveling (Slides 13-15), and the Minimum Moment Method (Slides 16-20). The reference chapter contains closely analogous resource-cap examples.

Coverage conclusion

The deck fully covers all repeated definition and theory questions. The two 2024 application questions are best treated as direct procedural matches: the charts and algorithms are present, but the examiner supplies a new interpretation task and a new nine-activity dataset.

3. Repeated Question Patterns

The topic became more applied over time: early papers asked for definitions and conceptual problems; 2024 required chart interpretation and a complete resource-constrained leveling exercise.

#	Pattern	Past questions	Slides	Answer strategy
1	Resource loading + associated problems	2017 Q6(a); 2022 Q8(a); first part of 2021 Q8(a)	6-8	Memorize the definition, explain why CPM/PERT is incomplete without resource availability, then distinguish time-limited and resource-limited cases.
2	Resource leveling + smoother distribution + simple network	2019 Q8(a); 2023 Q6(a); second part of 2021 Q8(a)	10-15	Use a network, an initial load profile and a shifted noncritical activity to show how peaks are reduced without changing the project duration.
3	Reading a resource-loading chart	2024 Q5(a)	9-13	High bars above capacity indicate overallocation; low bars, gaps or unused capacity indicate underutilization. Compare the profile with the available-resource limit.
4	Allocate and level under a resource cap	2024 Q5(c)	10-20	Perform CPM first, protect the critical path, move only activities with float, recalculate daily demand and select the smoother feasible schedule.

Year coverage

Year	Links	Marks	Main coverage
2016	0	0	No direct hit
2017	1	15	Resource loading definition and problems
2018	0	0	No direct hit
2019	1	12	Resource leveling and smoother distribution
2020	0	0	No direct hit
2021	1	12	Combined resource loading + leveling question
2022	1	12	Resource loading definition and problems repeated
2023	1	12	Resource leveling question repeated
2024	2	23	Chart interpretation + full numerical leveling

4. Slide-to-Question Reverse Index

Use this page while studying the files: each slide block shows which examination questions it can answer or support.

Slides	Content	Mapped questions	Priority
1-2	Title and reference book	No direct question	Low
3-5	Resources (4M) and loading = requirements	Background for 2017 Q6(a), 2021 Q8(a), 2022 Q8(a)	Medium
6	Definition, purpose and time-based resource demand	2017 Q6(a); 2021 Q8(a); 2022 Q8(a); supports 2024 Q5(a)	Critical
7-8	Allocation problem; time-limited vs resource-limited	2017 Q6(a); 2022 Q8(a); supports 2024 Q5(c)	Very high
9	Resource usage calendar	Supports 2024 Q5(a)	Medium
10-11	AOA network with resource use and slack	2019 Q8(a); 2021 Q8(a); 2023 Q6(a); 2024 Q5(c)	Very high
12	Load diagrams for Resources A and B	2019 Q8(a); 2021 Q8(a); 2023 Q6(a); 2024 Q5(a); 2024 Q5(c)	Critical
13	Definition and purpose of resource leveling	2019 Q8(a); 2021 Q8(a); 2023 Q6(a); supports 2024 Q5(c)	Critical
14-15	Advantages; network before and after leveling	2019 Q8(a); 2021 Q8(a); 2023 Q6(a); supports 2024 Q5(c)	Critical
16-20	Minimum Moment Method: formula, example, pros/limits and steps	Direct method for 2024 Q5(c); likely future numerical area	Critical
21	Activity EF/LF visual	Supporting visual only; use the reference chapter for complete worked details	Low
22	Thank-you slide	No exam content	None

Reference-file page map

Ref. PDF pages	Best use
3	Definitions of resource allocation/loading and resource leveling; why resources are leveled.
7-10	Minimum-moment introduction and Example 6.2: network, initial overallocation, use of float and improved profile.
11-19	Example 6.3: multiple resource limits, forward leveling, interruptible activities and resource-driven durations.
20-22	Advanced discussion: logic plus resource constraints, forward/backward leveling and schedule-flexibility options.
25	Chapter exercises that closely match the repeated definition questions.

5. 2024 Q5(c): Worked Verification and Slide Route

This bonus section is calculated from the activity table in the 2024 paper. It is not copied from the lecture. It shows one valid leveled schedule that respects the 10-labour limit and keeps the original critical-path duration.

Step	Action	Use	Result
1	Draw the network and run CPM	Slides 10-11	Critical path: A-B-C-F-G-I; project duration = 15 days.
2	Prepare the early-start resource profile	Slide 12	The early schedule peaks at 14 labourers on Days 6-7 and 11 labourers on Days 8-9.
3	Identify movable activities	Slides 13-16	D has 5 days of total float; the E-H branch has 4 days of float. Critical-path activities must remain fixed.
4	Shift within float and recalculate	Slides 16-20	Keep D on Days 6-9; move E to Days 10-11 and H to Days 12-14.
5	Verify the result	Slides 12, 18-20	Maximum daily demand becomes 10, and I still starts on Day 15. No critical-path increase occurs.

Activity schedule

Activity	Early schedule	One valid leveled schedule	Comment
A	1-2	1-2	Critical
B	3-5	3-5	Critical
C	6-9	6-9	Critical
D	6-9	6-9	Noncritical; retained
E	6-7	10-11	Shifted 4 days
F	10-11	10-11	Critical
G	12-14	12-14	Critical
H	8-10	12-14	Follows shifted E
I	15	15	Critical finish

Daily resource check

Profile	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Early	2	2	1	1	1	14	14	11	11	3	2	2	2	2	1
Leveled	2	2	1	1	1	10	10	10	10	6	6	3	3	3	1

Minimum-moment check

Early profile: $M = \sum(R^2) = 671$. Leveled profile: $M = 511$. The lower value confirms a smoother distribution, while the maximum demand is reduced from 14 to 10.

6. Exam-focused Study Plan

Recommended order for revising this lecture before the term final:

Order	Study source	Action
Step 1	Slides 6-8	Learn the exact definition of resource loading and the two allocation-problem categories. Prepare a 12-15 mark answer.
Step 2	Slides 10-15	Practice drawing the network, resource-load profile and a before/after leveling diagram. This answers the repeated 2019/2021/2023 theory pattern.
Step 3	Slides 16-20	Memorize the Minimum Moment Method: identify float, shift noncritical work, compute $M = \sum(R^2)$, and choose the lower-moment schedule.
Step 4	Reference 8-19	Study the worked resource-cap examples, especially how float removes a spike and how a resource limit can affect project duration.
Step 5	2024 Q5(c)	Rework the nine-activity problem without looking at the verification page. Confirm maximum demand ≤ 10 and unchanged critical-path duration.

Important limitations and updates

Issue	What to remember
Underutilization wording	Slide 12 shows the chart but does not explicitly define 'underutilization'. Explain it as resource demand below available capacity, visible through low bars or unused periods.
2024 numerical data	The exact nine-activity dataset is not in either source. The sources teach the method; the worked verification in this report is newly calculated from the examination table.
Slide 21	The EF/LF activity visual is incomplete as a standalone explanation. Use Slides 10-20 and reference pages 8-19 instead.
Reference file	5.1 is a textbook chapter, not a lecture slide set. Keep its PDF page references separate from the 5.0 slide numbers.
Terminology	The files use both 'resource allocation' and 'resource loading'. The reference chapter explicitly treats resource allocation as resource loading; use the examiner's wording in the answer.

Bottom line

For maximum return, prioritize Slides 6-8, 10-15 and 16-20, then use reference pages 8-19 for worked practice. These blocks cover all 86 mapped marks and every repeated question pattern in this lecture.